EPSR-test-ex23 xwts: X-ray form-factor weights

Program(s): xwts | chained passes: 1 | verification: ALL IDENTICAL (1 pass)

What this case tests

Exercises `xwts`, the keyword-driven program that prepares the `.wts` file for an X-ray data set: for every atom type it looks up the analytic atomic form factor and stores the coefficients together with the normalisation and the modified-atomic-form-factor (MAFF) parameters that `epsr` uses to weight the partial structure factors of an X-ray measurement [1,2].

Code exercised

Fortran (original, fortran_src/):

- `xwts.f`: `xwts` (component loop, prompt for unknown symbols, writing `<run>.wts`), `form_fact` (search of `f0_WaasKirf.dat`)
- `process_symbols.f`: `check_symbol` (`symbol_setup.eq.'xwts'` branch: `fnameato` defaults, `output`/`normtot` clamped); `setup_symbols_generic.f`: `setup_symbols_xwts`
- `read_and_save_files.f`: `read_input_file`, `read_ato`, `save_files_others`; `ato_read_and_save.f`: `minitc`

Rust (port, rust/epsr-core/src/):

- `tools/xwts.rs`: `run`, `form_fact`, `check_symbol_extra`
- `symbols.rs`: `read_input_file`, `save_inp_file`; `symtab_tools.rs`: table `XWTS`; `ato.rs`: `read_ato`, `minitc`

Physical system

The EPSR-test mixture of 200 DMSO and 400 water molecules (3200 atoms; atom types S, C, O, H, O1, H1) seen by X-rays. The form factor of each type is the Waasmaier-Kirfel five-Gaussian fit f0(s) = sum a_i exp(-b_i s²) + c [3]; each record of the `.wts` file holds c, the five (a_i, b_i) pairs and the MAFF q and delta values. `Xray1` uses neutral-atom factors with the defaults; `Xray2` uses per-molecule output, `normtot 1`, `iafformaff 2` (MAFFs used for the normalisation) and the ionic/valence factors `Cval`, `O1-`, `O2-` and `Hh+`, with MAFF parameters 0.5/0.7 on O1 and 0.3/0.45 on H1.

Inputs

`Xray{1,2,3}.XWTS.dat` written for this case (`Xray1` contains only `fnameato`); `f0_WaasKirf.dat` and `200DMSO400Water.ato` copied from EPSR-test. `xwts3.stdin` answers the prompts for the unknown symbols `Zz` and `Xx`: first the invalid `Qq`, then `S` and `Hh+`.

How the accuracy was verified

`commands.txt` (3 program calls, run once) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; no masking except where noted below). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in 1 pass.

One pass of three `xwts` calls; compared files: `Xray{1,2,3}.XWTS.wts` (format `11(1x,f11.6)` with format reversion) and the rewritten `Xray1`/`Xray3.XWTS.dat`. The reference shows the clamping: `Xray3.XWTS.wts` starts with `1 2 1` (`output 0` -> per atom, `normtot 5` -> 2, also written back to `Xray3.XWTS.dat`), `Xray2` with `2 1 2`; the C record of `Xray3` reproduces the tabulated Waasmaier-Kirfel coefficients of C (c = 4.297983, a1 = 2.657506, b1 = 14.780758).

Notes

`f0_WaasKirf.dat` is read from the current working directory, as in the Fortran.

References

[1] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[2] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[3] D. Waasmaier and A. Kirfel, "New analytical scattering-factor functions for free atoms and ions", Acta Cryst. A51, 416-431 (1995).
[4] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.